

CHAINSCOPE

Supply-chain answers you can trace to the source.

Governed conversational analytics for Snowflake CoCo CLI Hackathon - Problem Statement 05

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Synthetic demonstration data | Public repository and live replay included

A late shipment is not yet an explanation.

Operations teams need the affected commitment, approved metric and evidence in one governed answer.

Pain today

Supplier, inventory and customer-risk signals live in separate views.

Free-form assistants can mix definitions, invent joins or disclose commercial values.

Target users

Supply planners
Procurement teams
Plant operations
Commercial analysts
Governance and audit teams

Result

Trace an inbound problem to specific open demand.

Return deterministic metrics, approved SQL and exact source records - or refuse the question.

Ask -> govern -> calculate -> cite.

Seven business questions are mapped to explicit metric contracts and source evidence.

The screenshot displays the ChainScope application interface. The left sidebar contains navigation links: Overview, Ask copilot (active), Supply network, Metric catalog, and Audit trail. The main content area is titled 'Ask the supply chain' and shows a list of approved questions. The question 'Which customer orders are at risk?' is selected, leading to a results page titled 'Customer orders at risk' showing 4 orders. The results are presented in a table with columns: Order, Customer, Plant, Due date, Open orders, and Risk driver. Below the table, there is a section for 'Metric definitions & approved SQL' showing a checkmark and a link to 'All direct source records'.

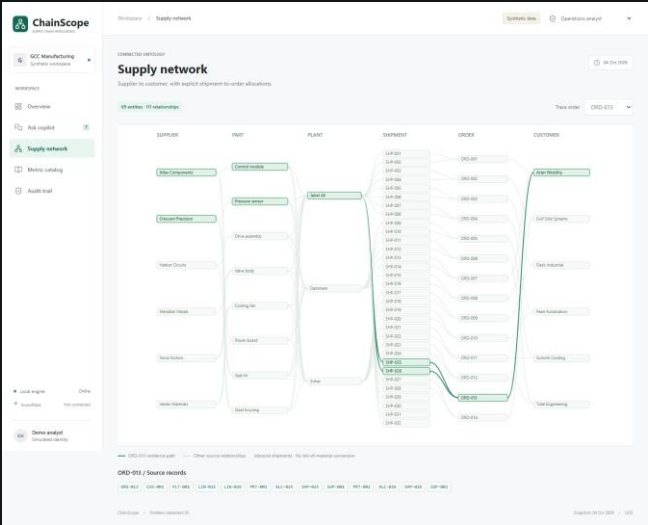
Order	Customer	Plant	Due date	Open orders	Risk driver
ORD-005	Summit Coating	Darmstadt	2028-10-02	70	Part due
ORD-006	Total Engineering	Darmstadt	2028-10-02	100	Part due
ORD-013	Aster Mobility	Jahad AI	2028-10-10	80	Late allocated supply
ORD-014	Gulf Star Systems	Jahad AI	2028-10-11	110	Late allocated supply

Live path

1. User selects an approved question
2. Role and field policy are checked before execution
3. Shared SQL metric runs at a defined grain
4. Result returns source IDs and records
5. Decision is written to the session audit trail

One ontology, shared metric contracts.

The local app proves behavior; Snowflake assets define the account deployment path.



Data layer

Suppliers -> parts -> plants -> shipments -> order lines -> customers

Governance layer

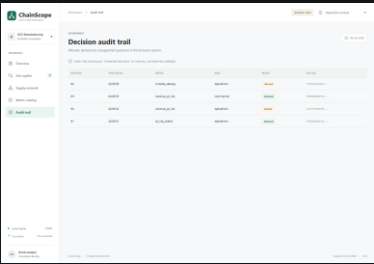
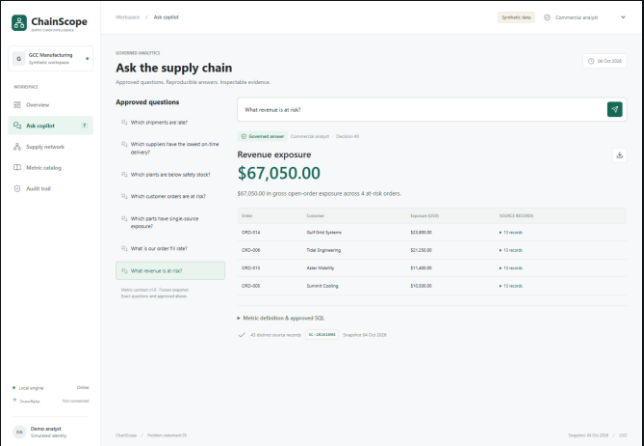
Question allowlist, role checks, redaction, source citations, audit chain

Snowflake path

Load SQL, shared views, native Semantic Views, object grants, Cortex Analyst request and CoCo review prompts

The right answer for the right role.

Commercial exposure is calculated from at-risk lines and denied before query for operations users.



Operations: denied | Commercial: USD 67,050 gross open-order exposure | Unsupported predictions: refused

Built to be inspected, extended and deployed.

Every figure below is from the reproducible synthetic snapshot - not a claimed client outcome.

56 tests passed

Metrics, integrity, API, security and static replay.

58 browser checks

29 local + 29 static; desktop and mobile.

7 governed metrics

Late supply, risk, inventory, fill, sourcing and exposure.

163 records

11 tables, 32 shipments, 14 orders.

Measured demo outputs

5 overdue inbound shipments
4 at-risk customer orders
55.2% supplier on-time delivery
41.6% unit fill

Reusable path

Static public replay + local app + CLI
Snowflake Semantic Views and least-privilege grants
Deterministic synthetic-data generator
GitHub-ready source and documentation